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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

V Semester B.E. Regular / Supplementary Examination February/ March 2026

COMMON TO CD / CS / CY / IS

Artificial Intelligence and Machine Learning

Time : 3 Hours

Maximum Marks : 100

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	Explain how a plateau differs from a ridge in hill-climbing search.	02	1	2
1.2	What is meant by informed search strategy? Give examples.	02	1	1
1.3	How does the size of a training set influence the risk of overfitting?	02	1	2
1.4	Differentiate supervised learning and unsupervised learning.	02	1	2
1.5	What are the two ways in which the prior probability can be obtained ?	02	1	2
1.6	A nearest neighbor classifier represents each example as a data point in a *d*-dimensional space. What does "d" represent here ?	01	1	2
1.7	State Baye's theorem	01	1	1
1.8	State one limitation of random initialization in K-Means clustering.	02	1	2
1.9	What is the role of cluster prototypes in compressing AI/ML datasets?	02	3	2
1.10	List any 4 typical AI problems.	02	2	2
1.11	What is an "uninformed search" ? Give an example.	02	1	2

Part B

Question No	Question	M	CO	BT
2a	Draw the schematic diagram indicating Structure of Model-based Reflex Agent and discuss the characteristics considering the Scenario of Automated Taxi. Write the Agent program for Model-based Reflex Agent.	08	2	3
2b	Write the Pseudocode for the recursive implementation of depth-limited tree search algorithm. Illustrate the same with suitable example.	04	1	3

2c	Differentiate between Uninformed and Informed search strategies and give two examples for each type.	04	1	2
3a	Explain how simulated annealing overcomes the limitations of hill-climbing search. Discuss the role of temperature and cooling schedule.	08	1	2
3b	Differentiate between A* Search, Minimax algorithm, and Alpha–Beta Pruning.	08	1	4
	OR			
4a	Explain why game-playing problems cannot be solved using BFS or DFS alone. Demonstrate with the help of an example	08	1	4
4b	Explain how a game can be formulated as a search problem. List and explain its components.	04	1	2
4c	Describe the working principle of simulated annealing. How does it overcome the limitation of hill-climbing search?	04	1	2
5a	Consider sample dataset given below for vertebrate classification. Discuss how attributes and class labels are used to build a classification model. Discuss the importance of selecting relevant attributes.	08	2	3

Body Temperature	Skin Cover	Gives Birth	Class Label
Warm-blooded	Hair	Yes	Mammal
Cold-blooded	Scales	No	Reptile
Cold-blooded	Scales	No	Fish
Warm-blooded	Feathers	No	Bird
Cold-blooded	None	No	Amphibian

5b	Using the following patient dataset with attributes *Fever*, *Lab Test Result*, and *Cough*, construct a decision tree to classify patients as *Infected* or *Not Infected*. Explain the working and how the model arrives at a decision in classifying a new instance.	08	2	3

Patient ID	Fever	Lab Test Result	Cough	Class Label
P1	Yes	Positive	Yes	Infected
P2	Yes	Positive	Yes	Infected
P3	Yes	Negative	Yes	Infected
P4	No	Negative	No	Not Infected
P5	No	Negative	No	Not Infected
P6	Yes	Positive	No	Infected
P7	No	Positive	Yes	Infected
P8	No	Negative	No	Not Infected

	OR			
6a	Describe binary and multiclass classification with suitable examples. Suggest appropriate learning algorithm for each case and justify the same	08	3	4

6b	Differentiate between internal nodes and leaf nodes in a decision tree.	04	2	2
6c	How does "Incorporating Model Complexity" act as a mechanism for controlling the Bias-Variance tradeoff?	04	2	4
7a	Consider the below given dataset. Name, Age, Gender, Sports Ajay, 32, 0, Football Mark, 40, 0, Neither Sara, 16, 1, Cricket Zaira, 34, 1, Cricket Sachin, 55, 0, Neither Rahul, 40, 0, Cricket Pooja, 20, 1, Neither Smith, 15, 0, Cricket Laxmi, 55, 1, Football Michael, 15, 0, Football Use K-NN algorithm to find the class of sports that Angelina likes. Her age is given as 5 and assume K=3 Also, explain the algorithm	08	1	3
7b	Write the pseudocode and explain the technique of bootstrap aggregation with the help of an example . What are its advantages	08	1	3
	OR			
8a	What are the two ways to estimate the class-conditional probabilities for continuous attributes ?	04	2	1
8b	Explain the characteristics of Na¨ıve Bayes Classifiers ?	08	1	2
8c	Give the pseudocode for the *k*-nearest neighbor classifier. Suggest a method to reduce the impact of k on classification	04	2	2
9a	Describe how empty clusters, outliers, centroid updates are handled in k-means clustering algorithm	06	2	2
9b	The Bisecting K-means algorithm is a top-down hierarchical approach. How does it improve upon the stability and quality of results compared to a single run of standard K-means?	04	2	1
9c	Distinguish the following types of clusterings: i) Hierarchical versus Partitional ii) Exclusive versus Overlapping versus Fuzzy iii)Complete versus Partial	06	2	2
	OR			
10a	Explain the internal measures used for evaluating clustering validity. Illustrate with suitable examples.	08	2	2
10b	Explain the different types of data used in clustering analysis with examples.	08	1	2